Activities and Methodology

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Project Title: Finite State Automata and Regular Expression Simulator

The planned activities of the project will be a review of existing solutions, a literature review, requirement analysis, an initial design phase, implementation and then finally an evaluation.

Reviewing existing solutions will involve searching the web for different applications that implement finite state automata simulators, regardless of whether regular expressions were implemented or not. While the project is related to regular expressions, it is more related to regular expressions in terms of finite state automata, therefore I will focus more on the automata aspect. The review will involve looking at what features are present and not, as well as reviewing if there are any limitations in the features present. The literature review involves looking at the textbooks in the reading lists of COMP2210 to get insights into the logic of algorithms that need to be implemented. This review will also help find restrictions that need to be placed into the project. For example, I will need to limit the sizes of NFAs when doing subset construction to prevent the created DFAs from being too large.

For requirement analysis, I shall be doing stakeholder analysis, creating UI mock-ups, creating user stories and creating use case diagrams. Stakeholder analysis will involve identifying the stakeholders and creating personas and scenarios to help see the users' needs. I will use Figma to create mock-ups so I can visualise how the solution will look and function without having to program. I will also use the personas and scenarios to help create user stories as functional requirements of the project. These user stories will be prioritised using MoSCoW so that the most important requirements are met first. Lucidchart will be used to create use case diagrams to help visualise how the user will interact with the system to achieve these user stories.

In the initial design phase, I will be using Lucidchart to create UML diagrams such as class diagrams and activity diagrams and sequence diagrams. These diagrams will be made to help visualise the structure of the system and how the different components of the system will interact with each other.

For the implementation, I will be following the agile methodology to allow changing requirements and to get consistent feedback to ensure the project meets users' needs. The implementation phase will be split into 3 sprints, where each sprint will deliver a prototype of the solution and the final sprint will deliver the finished product. Feedback given to the prototypes at the end of sprints and in weekly supervisor meetings will be used to refine the prototypes as necessary. Unit tests and acceptance tests will be done during implementation to ensure the user stories are correctly implemented. The implementation will be written in Java and JavaFX as they are platform independent and I can use Maven to build and manage the Java project easier. IntelliJ or Visual Studio Code are choices for the IDE, but as I am using Maven, it will be best to use IntelliJ since Maven is built in. I will also be using GitLab for the version control.

Finally for the evaluation, a survey will be created using Microsoft Forms to ask students questions such as how they would rate the usability of the solution and whether there are features that are not implemented well. The results will be analysed using a mixture of quantitative and qualitative methods. For example, thematic analysis could be done on the qualitative questions and regression analysis could be done on the quantitative questions if I get enough responses in the survey.